

Problem 1:

$$24 : 36 = \frac{2}{3} \frac{4}{6}$$

$$GCD(24, 36) = 12$$

$$\text{Simplified num.} = 2 \times 12$$

$$\text{Simplified den.} = 3 \times 12$$

$$\text{Simplified fraction} = \frac{2 \times 12}{3 \times 12}$$

$$= \frac{2}{3} \times \frac{12}{12}$$

$$= \frac{2}{3}$$

$$\frac{2}{3} = 2 : 3$$

$$\begin{array}{r} 24 \overline{) 36} (1 \\ \underline{24} \\ 12 \\ 24 \\ \underline{24} \\ 0 \end{array}$$

Problem 2:

$$40 : 120 = \frac{40}{120}$$

$$\boxed{\begin{matrix} \text{B} & \text{e} & \text{c} & \text{a} & \text{u} & \text{s} & \text{e} \\ 2 & 2 & 0 & 2 & 5 & 5 & 5 \\ 1 & 2 & 0 & 2 & 3 & 3 & 3 \end{matrix}} \rightarrow$$

$$\text{GCD}(40, 120) = 40$$

$$\text{Simplified num.} = 1 \times 40$$

$$\text{Simplified den.} = 3 \times 40$$

$$\text{Simplified fraction} = \frac{1 \times 40}{3 \times 40}$$

$$= \frac{1}{3} \times \frac{40}{40}$$

$$= \frac{1}{3}$$

$$\text{Ratio} = 1 : 3$$

$$40 \overline{) 120} \begin{matrix} 3 \\ \underline{120} \\ 0 \end{matrix}$$

Problem 3:

$$\begin{aligned}\frac{7}{20} &= \left(\frac{7}{20} \times \frac{100}{100} \right) \% \\ &= \left(7 \times \frac{5}{100} \right) \% \\ &= (7 \times 5) \% \\ &= 35 \%\end{aligned}$$

$$\frac{7}{20} = x\%$$

$$\Rightarrow \frac{7}{20} = \frac{x}{100}$$

$$\Rightarrow \frac{7}{20} \times 100 = \frac{x}{100} \times 100$$

$$\Rightarrow x = \frac{7}{20} \times 100 = 35$$

Problem 4:

$$\begin{aligned}
 12.5\% &= 12.5 \times \frac{1}{100} \\
 &= \frac{12.5}{100} = \frac{12.5}{100} \times \frac{10}{10} \\
 &= \frac{125}{1000}
 \end{aligned}$$

$$6 \text{ C D C } 125, 1000) = 125$$

$$125 = 125 \times 1$$

$$100 = 125 \times 8$$

$$\begin{aligned}
 \text{Simple fraction} &= \frac{125 \times 1}{125 \times 8} \\
 &= \frac{125}{1000} = \frac{1}{8}
 \end{aligned}$$

$$\begin{array}{r}
 125 \overline{) 1000} \\
 \underline{1000} \\
 0
 \end{array}$$

$$\begin{array}{r}
 \begin{array}{c} 2 \quad 4 \\ 1 \quad 2 \quad 5 \end{array} \\
 \times \quad \begin{array}{c} 1 \quad 2 \quad 5 \end{array} \\
 \hline
 1 \quad 0 \quad 0 \quad 0
 \end{array}$$

problem 5:

Total marbles = 50

Red marbles = 15

Non-red marbles = $50 - 15$
= 35

Ratio = 15 : 35

Simplify your ratio = 3:7

Problem 6:

Original price = x

Given 20% of $x = \$12$

~~$$20 \times \frac{1}{100} \times x = \$12$$~~

$$\frac{20}{100} \times x = \$12$$

$$\frac{100}{20} \times \frac{20}{100} \times x = \$\left(\frac{100}{20}\right) \times 12$$

$$1 \times x = \$ (5 \times 12)$$

$$x = \$60$$



Problem 7:

5 shirts cost = \$75

12 shirts cost = \$x

~~No. 4~~ ~~shirts increases~~ → cost increases,
direct proportion

$$\frac{x}{12} = k$$

$$\frac{\$75}{5} = k$$

$$\Rightarrow \frac{x}{12} = \underline{\$15}$$

$$\Rightarrow \frac{x}{12} = \$15$$

$$\Rightarrow \frac{x}{12} \times 12 = \$15 \times 12$$

$$\Rightarrow x = \$180$$



problem 8: Direct proportion: $1:50,000 = 8:x$

Actual distance $1 \text{ cm} =$
 $1:50,000$

Actual distance $8 \text{ cm} =$
 $8:50,000$
 $= 8:400,000$

Map Scale = $1:50,000$

$$\Rightarrow \frac{\text{map distance}}{\text{actual distance}} = \frac{1}{50,000}$$

Both num, den in same units $\frac{\text{cm}}{\text{cm}} = \frac{\text{m}}{\text{m}} = \frac{\text{km}}{\text{km}} = 1$

So you don't have units for the ratio

If map distance = 8 cm

$$\Rightarrow \frac{8 \text{ cm}}{\text{act}} = \frac{1}{50,000} \Rightarrow \text{act} = 50,000 \times 8 \text{ cm}$$
$$= 50,000 \times 8 \times \frac{1}{100} \text{ m} = 50,000 \times 8 \times \frac{1}{100} \times \frac{1}{1000} \text{ km} = 4 \text{ km}$$

Problem 9: Direct proportion - $4:200 = 14:x$

Flour for 4 people = 200g

Flour 14 people = x

Ratio = people : flour

$$\frac{4}{200} = \frac{14}{x}$$

$$\Rightarrow \frac{4}{200} \times x = \frac{14}{x} \times x$$

$$\Rightarrow \frac{4}{200} \times x = 14$$

$$\Rightarrow \frac{4}{200} \times x \times \frac{200}{4} = 14 \times \frac{200}{4}$$

$$\Rightarrow x = 14 \times 50$$

$$\Rightarrow x = 700g$$

$$\frac{4p}{200g} = \frac{14p}{x}$$

$$\Rightarrow x = \frac{14p \times 200g}{4p} = 700g$$

problem 10: Direct proportion - $20:240 = x:600$

Petrol for 240 km = 20 litres

Petrol for 600 km = x litres

⇒ petrol $\propto$ distance

$$\frac{x}{600} = \frac{20}{240} = k$$

⇒
⇒
⇒ $\frac{\text{petrol}}{\text{distance}} = k$

$$600 \times \frac{x}{600} = 600 \times \frac{20}{240}$$

$$\frac{600}{600} \times x = 600 \times \frac{1}{12}$$

$$x = \frac{600}{12}$$

$$x = 50 \text{ litres}$$



problem 11: Direct proportion - $y:x=5$

$$x \times 5 = y$$

$$7 \times 5 = y$$

$$y = 35$$

$$\frac{y}{x} = k$$

$$\frac{y}{x} = 5$$

$$\frac{y}{7} = \frac{10}{2}$$

$$y = 5 \times 7$$
$$= 35$$

Step
by
step

$$y \propto x \Rightarrow \frac{y}{x} = k$$
$$\Rightarrow \frac{10}{2} = \frac{y}{7}$$

$$\Rightarrow y = 5 \times 7 = 35$$

Problem 12: Direct proportion

Actual length of pole = 2 m

Shadow length of pole = 3.5 m

Building shadow length = 28 m

Building length = x

$$\text{Ratio} = \frac{\text{Actual length}}{\text{Shadow length}} = k$$

$$\Rightarrow \frac{x}{28}$$

$$= \frac{2}{3.5}$$

$$\Rightarrow \frac{x}{28} \times 28 = \frac{2}{3.5} \times 28$$

$$\Rightarrow x = \frac{2 \times 28}{3.5}$$

$$\Rightarrow x = \frac{56}{3.5}$$

$$= \frac{560}{35}$$

$$= 16 \text{ m}$$



$$\begin{array}{r} 16 \\ 35 \overline{) 560} \\ \underline{35} \\ 210 \\ \underline{210} \\ 000 \end{array}$$

$$\times \begin{array}{r} 35 \\ 210 \end{array}$$

Problem 13: Indirect proportion

Pipes $\times$ time = k no. of Pipes $\propto \frac{1}{\text{time}}$

Time for 6 pipes = 80 min

Time for 5 pipes = y

$$5 \times y = 6 \times 80$$

$$\frac{5 \times y}{5} = \frac{6 \times 80}{5}$$

$$y = 6 \times \frac{80}{5}$$

$$= 6 \times 16$$

$$= 96 \text{ min}$$



Problem 14:

Speed $\propto \frac{1}{\text{time}} \Rightarrow$

Speed $\propto \frac{1}{\text{time}} \Rightarrow$

Journey in 90 km / h = 4

Journey in 120 km / h = x

$$90 \times 4 = 120 \times x$$

$$\Rightarrow 360 = 120x$$

$$\Rightarrow \frac{360}{120} = \frac{120x}{120}$$

$$\Rightarrow 3 = x$$

$$\Rightarrow x = 3 \text{ hrs}$$

$$90 \frac{\text{km}}{\text{h}} \times 4 \text{ h} = 120 \frac{\text{km}}{\text{h}} \times x$$

$$\Rightarrow x = \frac{90 \times 4}{120} \text{ h}$$

$$= 3 \text{ h}$$

Problem 15:

Workers time = k

No. of workers + 1 day
= 10

No. of workers + 3 days
= x

$$3x = 12x + 10$$

$$\Rightarrow \frac{3}{3}x = \frac{12x}{3} + \frac{10}{3}$$

$$\Rightarrow x = 4x + \frac{10}{3}$$

$$\Rightarrow x = 4x + 10$$

$$\Rightarrow x = 40$$

See AI notes on
which work is
sequential / parallel

Problem 16:

$$x \times y = k$$

$$y \times x = 24$$

$$y \times 3 = 6 \times 4$$

$$\Rightarrow \underbrace{y \times 3}_{\text{}} = \frac{6 \times 4}{3}$$

$$\Rightarrow y = 8$$



Problem 17:

D O g s x d a y s = k

N o . o t d a g s t o r 5 d o g s
= 1 2 ←

N o . o t d a g s t o r 1 5 d o g s
= x ←

$$1 \ 5 \ x \ x = 1 \ 2 \ x \ 5$$

$$\Rightarrow \frac{1 \ 5 \ x \ x}{1 \ 5} = \frac{1 \ 2 \ x \ 5}{1 \ 5}$$

$$\Rightarrow x = \frac{6 \ 0}{1 \ 5}$$

$$\Rightarrow x = 4 \quad \checkmark$$

Problem 18: I need help.

$$\frac{\text{part 1}}{\text{part 2}} = \frac{4}{1} ; \text{part 1} + \text{part 2} = 200$$

$$\Rightarrow \text{part 1} = 4 \times \text{part 2} ; \Rightarrow 4 \times \text{part 2} + \text{part 2} = 200$$

$$\Rightarrow (4+1) \text{part 2} = 200$$

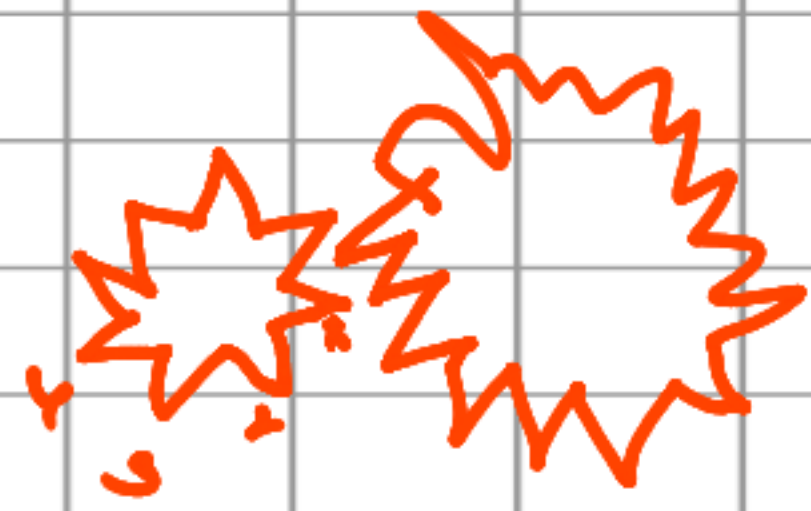
$$\Rightarrow 5 \times \text{part 2} = 200$$

$$\Rightarrow \text{part 2} = 40$$

$$\Rightarrow \text{part 1} = 4 \times \text{part 2} = 4 \times 40 = 160$$

$$15\% \text{ of part 1} = \frac{15}{100} \times 160 = 24$$

Problem 19: I need help.



Each revolution means all its teeth have rotated once and they are back in the initial position.

If first gear moves one tooth, second one also moves one tooth.

Total n. of teeth moved is constant

⇒ n. of teeth in the gear $\times$ no. of revolutions = k.

$$\Rightarrow 20 \text{ teeth} \times 100 \text{ rev} = 50 \text{ teeth} \times x \text{ rev}$$

$$\Rightarrow x = \frac{20 \times 100}{50} = 40 \text{ rev.}$$

Problem 20:

$$x \% =$$

$$b = 3$$

$$g = 5$$

$$s = 8$$

$$x \% = \frac{\cancel{8}}{s} \quad \frac{b}{s}$$

$$\frac{x}{100} = \frac{5}{8}$$

$$\frac{x}{100} \times 100 = \frac{5}{8} \times 100$$

$$x = \frac{500}{8}; \quad x = 62.5$$

$$\begin{array}{r} 62.5 \\ 8 \overline{) 500} \\ \underline{48} \\ 20 \\ \underline{16} \\ 40 \\ \underline{40} \\ 00 \end{array}$$

$$\frac{b}{g} = \frac{3}{5} \quad (1)$$

$$b + g = 800 \quad (2)$$

$$\frac{g}{800} = x \% = \frac{x}{100}$$

$$\Rightarrow x = \frac{g}{800} \times 100 = \frac{g}{8}$$

(3)

$$b = \frac{3}{5} \times g; \text{ (from (1))}$$

$$\Rightarrow \frac{3}{5} \times g + g = 800 \quad (\text{from (2)})$$

$$\Rightarrow \left(\frac{3}{5} + 1\right)g = 800$$

$$\Rightarrow \frac{8}{5} \times g = 800$$

$$\Rightarrow g = \frac{800 \times 5}{8} = 500$$

$$\text{From (3)} \quad x = \frac{g}{8} = \frac{500}{8} = 62.5$$